

Cloud Computing Overview and Concepts

Cloud overview, Characteristics and Benefits

Capex vs Opex Model

Cloud Service Models

Shared Responsibility Model

Cloud Deployment Models

Cloud Service Comparison: AWS, Azure, Google

Cloud What & Why?

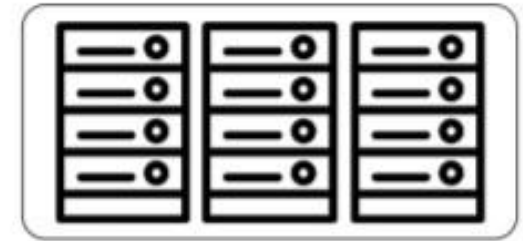
Traditional Architecture



Home or Garage



Office



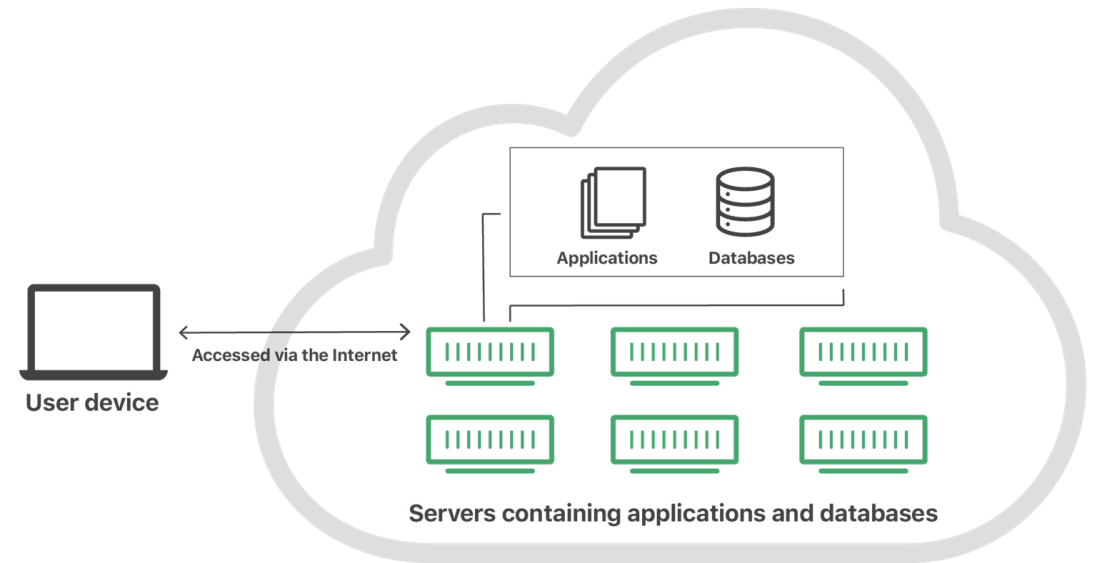
Data center

Problem With Traditional Arch.

- Pay for the rent for the data center
- Pay for power supply, cooling, maintenance
- Adding and replacing hardware takes time
- Scaling is limited
- Hire 24/7 team to monitor the infrastructure
- How to deal with disasters? (earthquake, power shutdown, fire)
- Can we externalize all this?

What is Cloud Computing?

- Cloud computing is the delivery of computing services: servers, storage, databases, networking, tools and software over the Internet.
- Cloud computing enables companies to consume a compute resource, such as a **servers, storage, networking** or an **application**, as a utility like water or electricity, rather than having to build and maintain computing infrastructures in house.



Cloud Characteristics



On-demand
self-service

No human
intervention
needed to
get
resources



Broad
network
access

Access
from
anywhere



Resource
pooling

Provider
shares
resources
to
customers



Rapid
elasticity

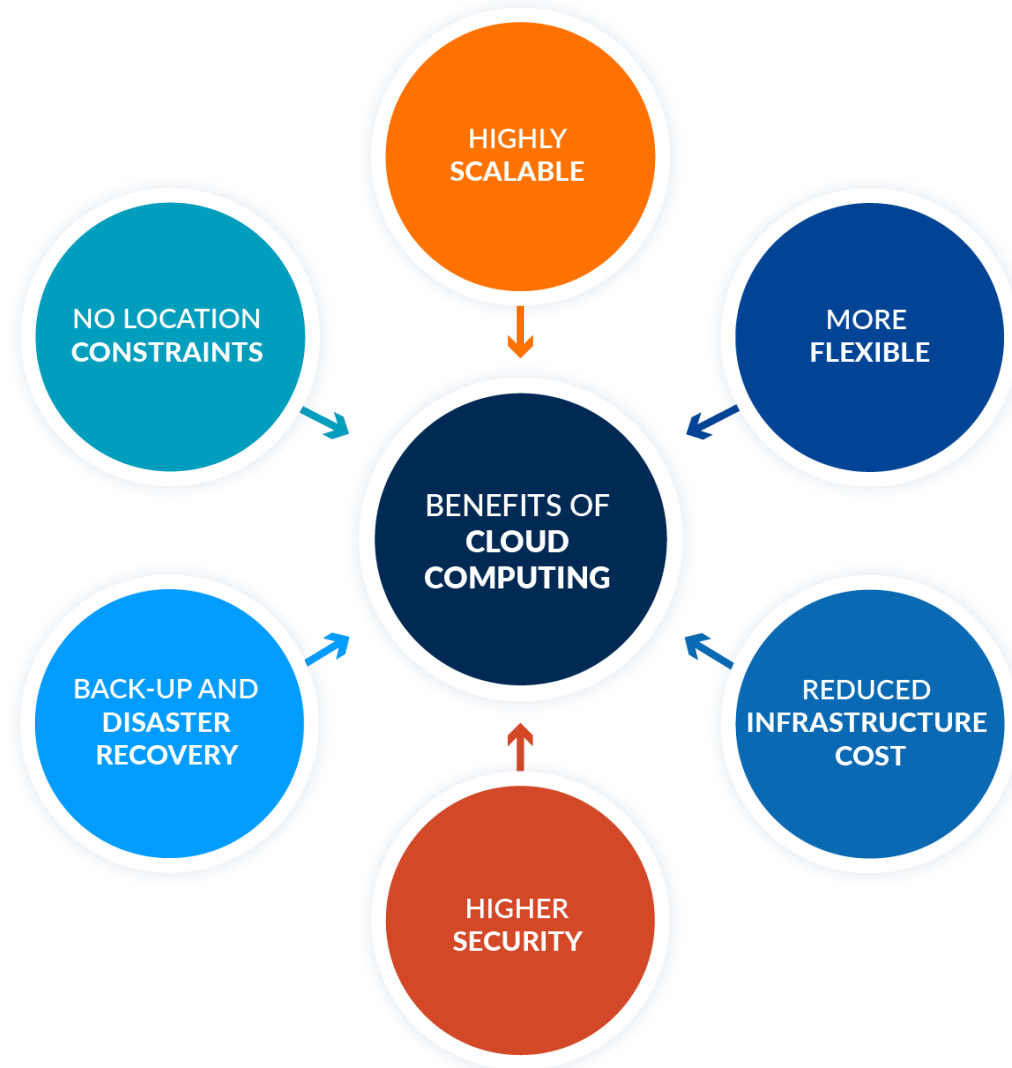
Get more
resources
quickly as
needed



Measured
service

Pay only
for what
you
consume

Why Cloud



High Availability

High Availability

Maintaining acceptable continuous performance despite temporary load fluctuations or failure in services, hardware, datacentres

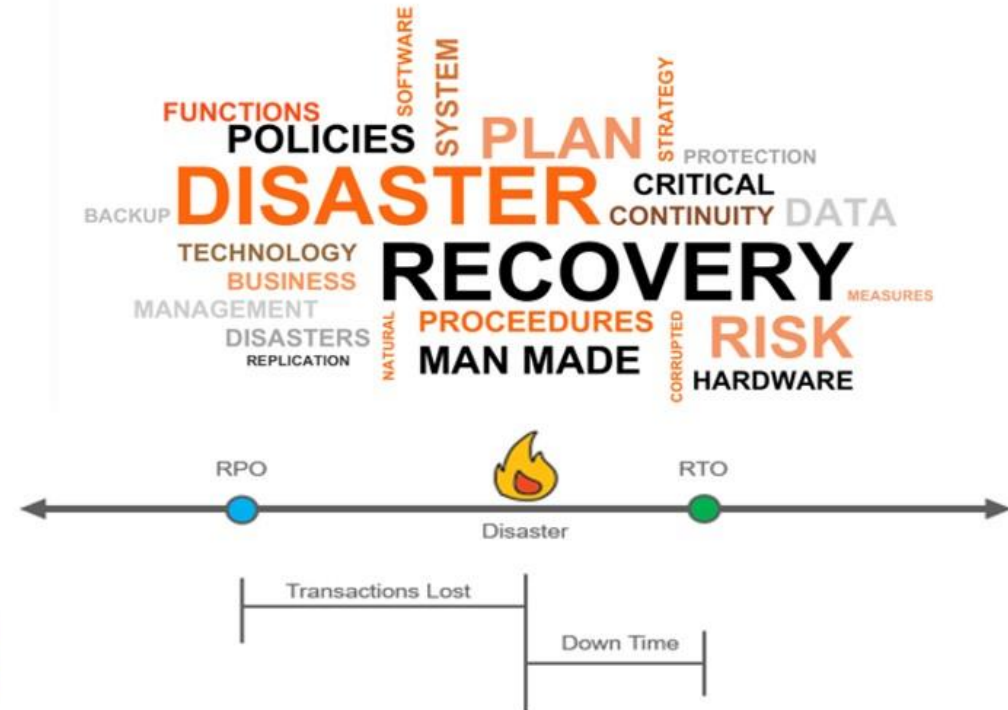
Data Centre Redundancies

- Power
- Cooling
- Networking



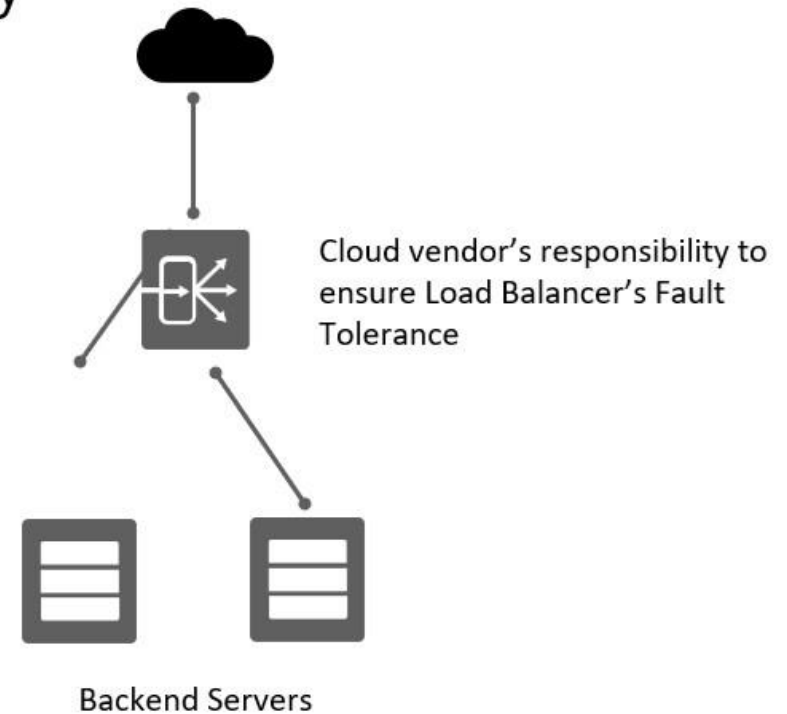
Disaster Recovery

- Disaster Recovery (DR) involves a set of policies, tools and procedures to enable the recovery or continuation of vital technology infrastructure and systems
- When designing Disaster Recovery consider
 - Recovery Point Objective (RPO)
 - Recovery Time Objective (RTO)
- Consider Time it takes to recover and amount of data you can afford to lose



Fault Tolerance

Systems ability to continue operations properly when one or more of its components fails.



Elasticity

Elasticity in cloud computing refers to the **ability to automatically scale up or down the resources** used by a cloud-based application in response to changes in demand. This can help to ensure that applications are always running with the resources they need, without wasting resources when they are not needed.

- Vertical & Horizontal Scaling
- Manual or Automatic ➤ Based on Workload
- Pay only what you use



Scalability

Ability to increase instant count or Size of existing resource based on Traffic Fluctuations, load etc.

Scale-out - Horizontal Scaling

- Upgrade the capacity of the app by increasing the number of host instances
- Ex: Having a Load Balancer where your app is hosted on multiple instances.

Scale-up - Vertical Scaling

- Upgrade the capacity of the host where the app is hosted
- Ex: Increase the RAM size from 1 cores to 4 cores.

SCALE-UP



SCALE-OUT



Elasticity vs Scalability



Cloud Scalability

Scalability is a strategic operation for adding computing resources to support demand. As workload grows (AKA "a good problem to have"), you want a system that can handle growth.

Cloud Elasticity

Cloud server elasticity represents more of a tactical approach to allocating computing resources. Elasticity provides the necessary resources required for the current workload but also scales up or down to handle peak utilization periods as well as off-peak loads.

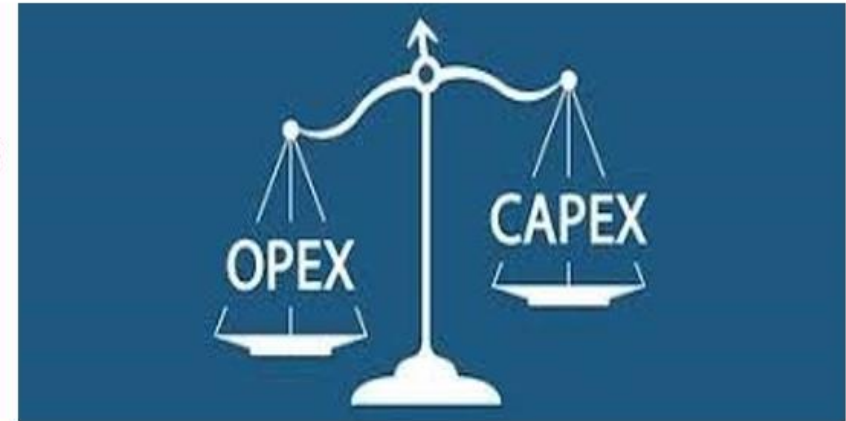
Capex vs Opex

Capex

- Upfront Investment
- Create benefit over longer period of time
- Usually non-recurring
- Related to Purchase or permanent assets

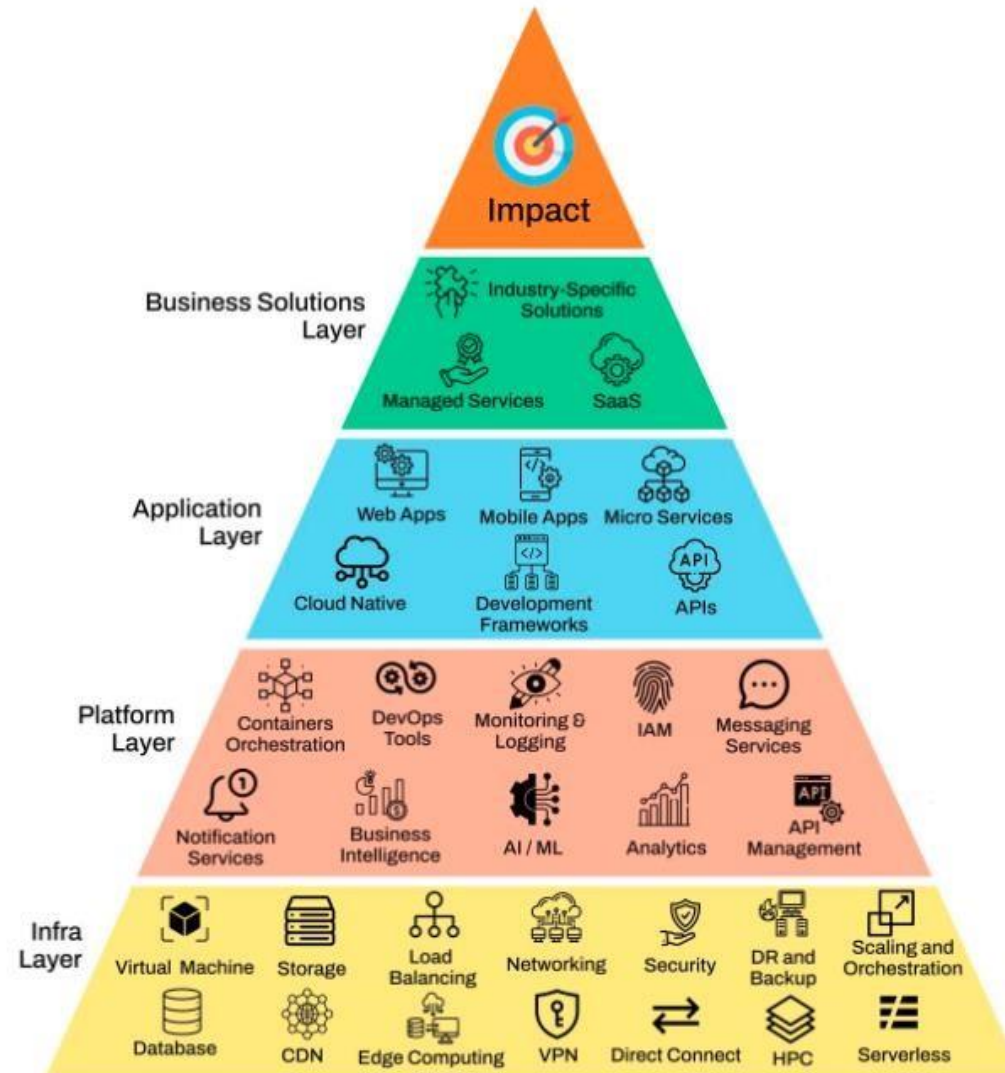
Opex

- No up-front Investment- Pay as you Go
- Creates Immediate benefits
- Recurring cost of doing business - Operation



Service Models IaaS | PaaS | SaaS

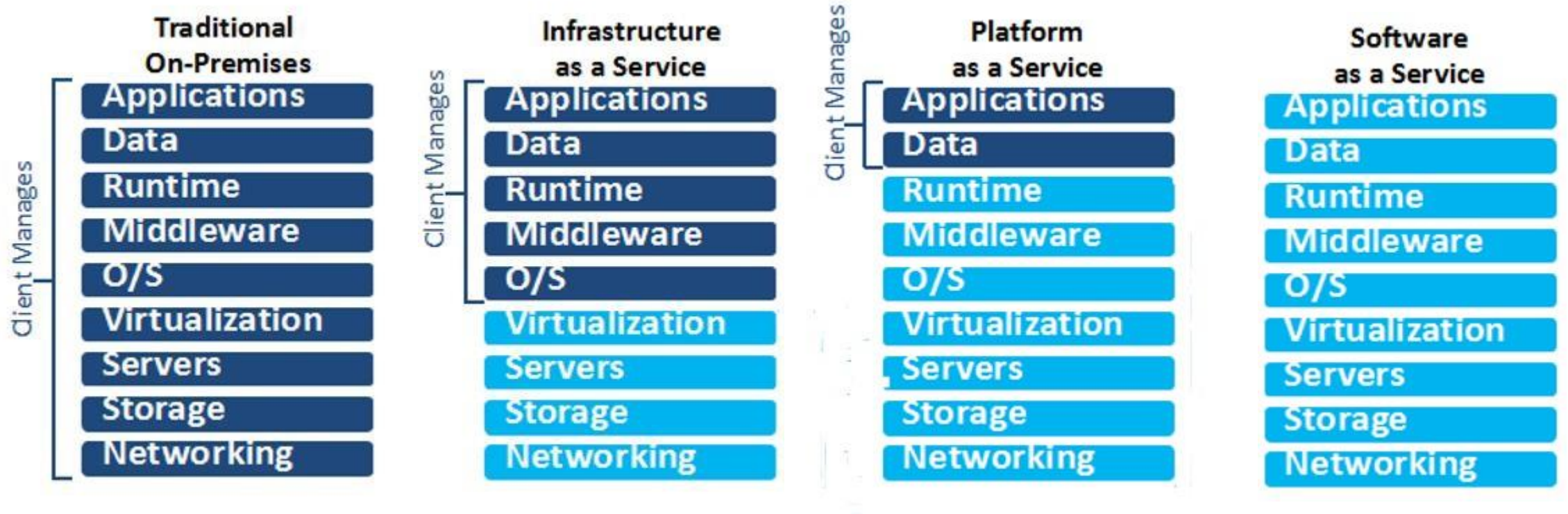
Cloud Service Pyramid



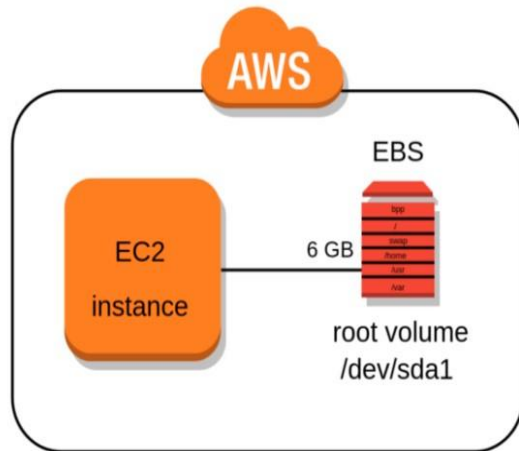
Cloud Service Models



Cloud Service Model



Example: IaaS (AWS)



Amazon Elastic Compute Cloud (Amazon **EC2**) provides scalable computing capacity in the Amazon Web Services (AWS) Cloud. Using Amazon EC2 eliminates your need to invest in hardware up front, so you can develop and deploy applications faster. You can use Amazon EC2 to launch as many or as few virtual servers as you need, configure security and networking, and manage storage. Amazon EC2 enables you to scale up or down to handle changes in requirements or spikes in popularity, reducing your need to forecast traffic.

Example: PaaS (AWS)

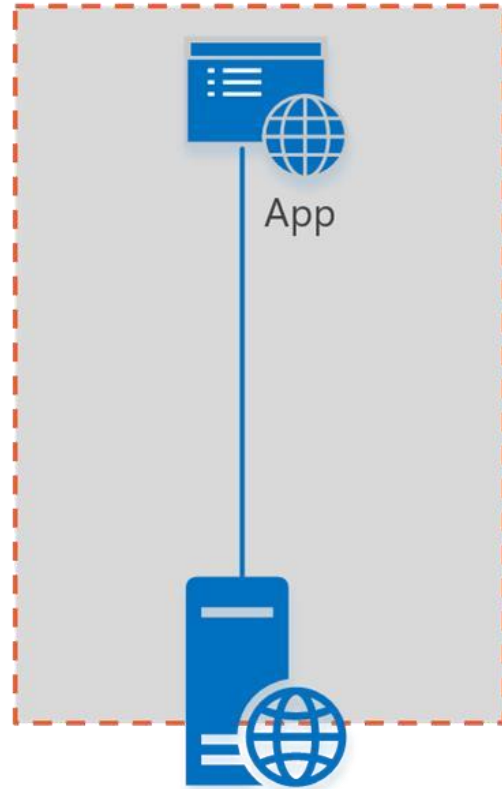


AWS Elastic Beanstalk is an easy-to-use service for deploying and scaling web applications and services developed with Java, .NET, PHP, Node.js, Python, Ruby, Go, and Docker on familiar servers such as Apache, Nginx, Passenger, and IIS.

You can simply upload your code and Elastic Beanstalk automatically handles the deployment, from capacity provisioning, load balancing, auto-scaling to application health monitoring. At the same time, you retain full control over the AWS resources powering your application and can access the underlying resources at any time.

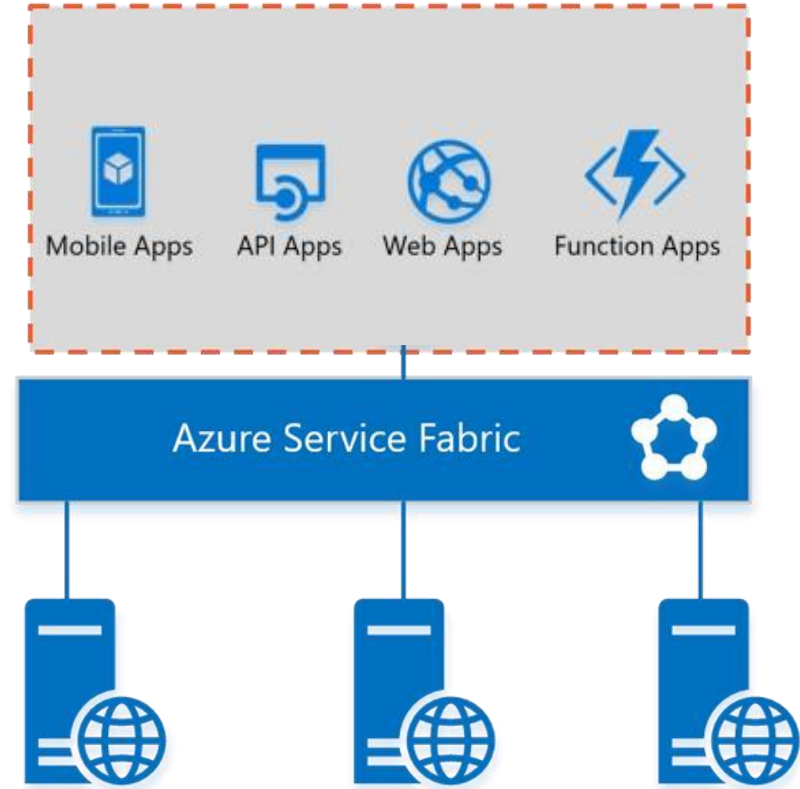
Example: IaaS & PaaS (Azure)

Traditional Webhosting



Server + OS + Webserver (IIS)

Hosting with Apps



Server + Azure OS + Webserver (IIS)

Example: IaaS & PaaS (Azure)

SQL Server on Azure Virtual Machines



Best for re-hosting and apps requiring OS-level access and control

Automated manageability features and OS-level access

Infrastructure as a Service

Azure SQL Managed Instance



Best for modernizing existing apps

Offers high compatibility with SQL Server and native VNET support

Platform as a Service

Azure SQL Database



Best for building new apps in the cloud

Pre-provisioned or serverless compute and Hyperscale storage to meet demanding workload requirements

Example: SaaS (Azure)

Hosting with Apps

- Google Workspace (formerly GSuite)
- Dropbox.
- Salesforce.
- Cisco WebEx.
- SAP Concur.
- GoToMeeting.



Word



Excel



PowerPoint



OneNote



Access



Publisher



Outlook



Lync



InfoPath

Example: IaaS (Google)

- Products like Compute Engine, Cloud Storage, Google VPC are all examples of IaaS



Google Cloud Storage



Google Compute Engine

Example: PaaS (Google)

- Products like Google Kubernetes Engine and Google App Engine come under Platform as a Service



Google Kubernetes Engine



App Engine

Example: SaaS (Google)

- All the offerings in Google Cloud Marketplace come under Software as a Service



Example: IaaS | PaaS | SaaS (Oracle)

Oracle SaaS



Sales
Cloud



Global Human
Resources Cloud



ERP
Cloud

Oracle IaaS



Compute



Object
Storage



Load
Balancing

Oracle PaaS



Database
Cloud



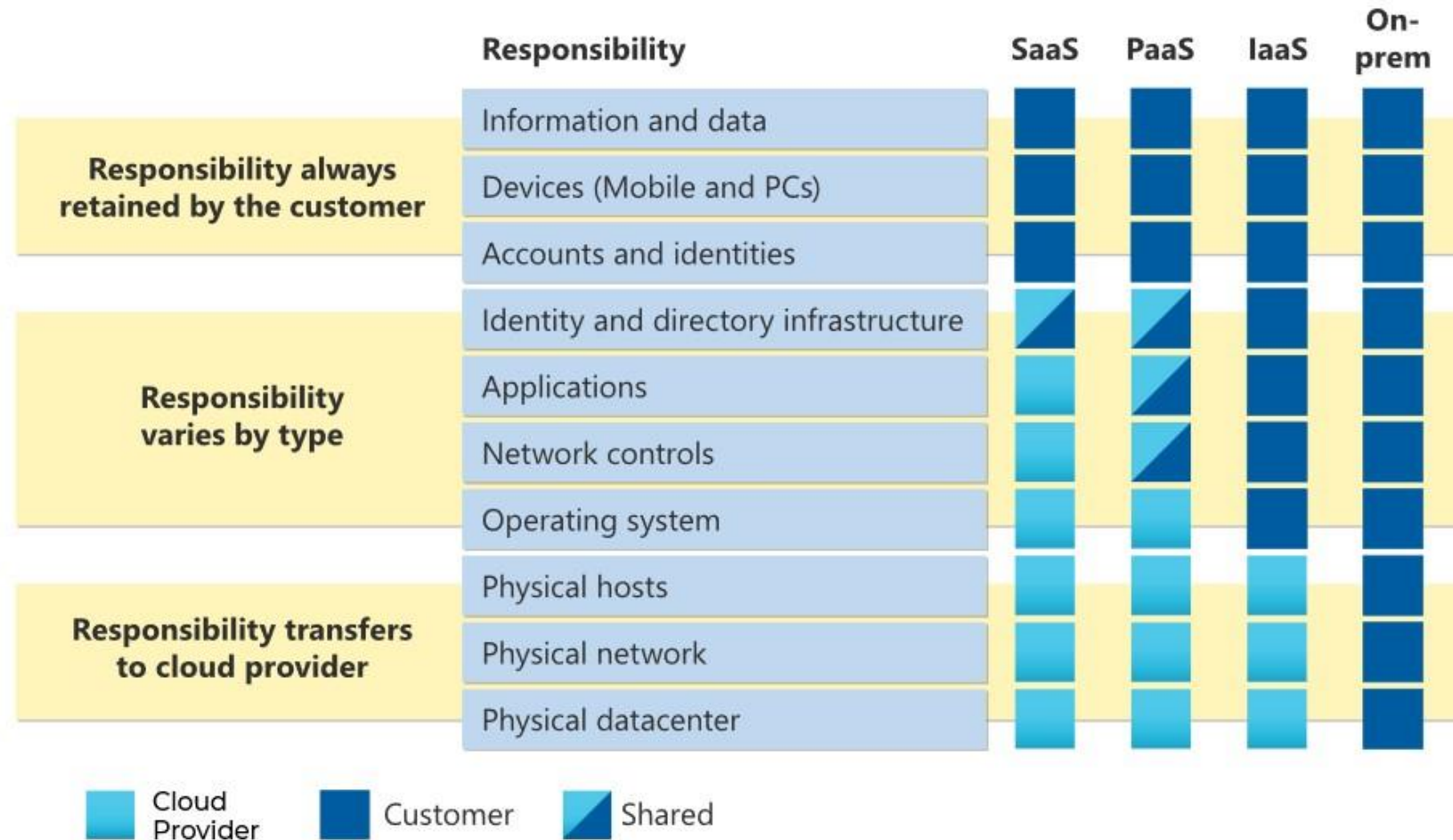
Content and
Experience Cloud



Java
Cloud

Shared Responsibility

Shared Responsibility Model



Cloud Deployment Model



PUBLIC CLOUD

- Offered by third-party providers
- Available to anyone over the public internet
- Scales quickly and convenient



HYBRID CLOUD

- Combination of both public & private cloud
- Shared security responsibility
- Helps maintain tighter controls over sensitive data & processes

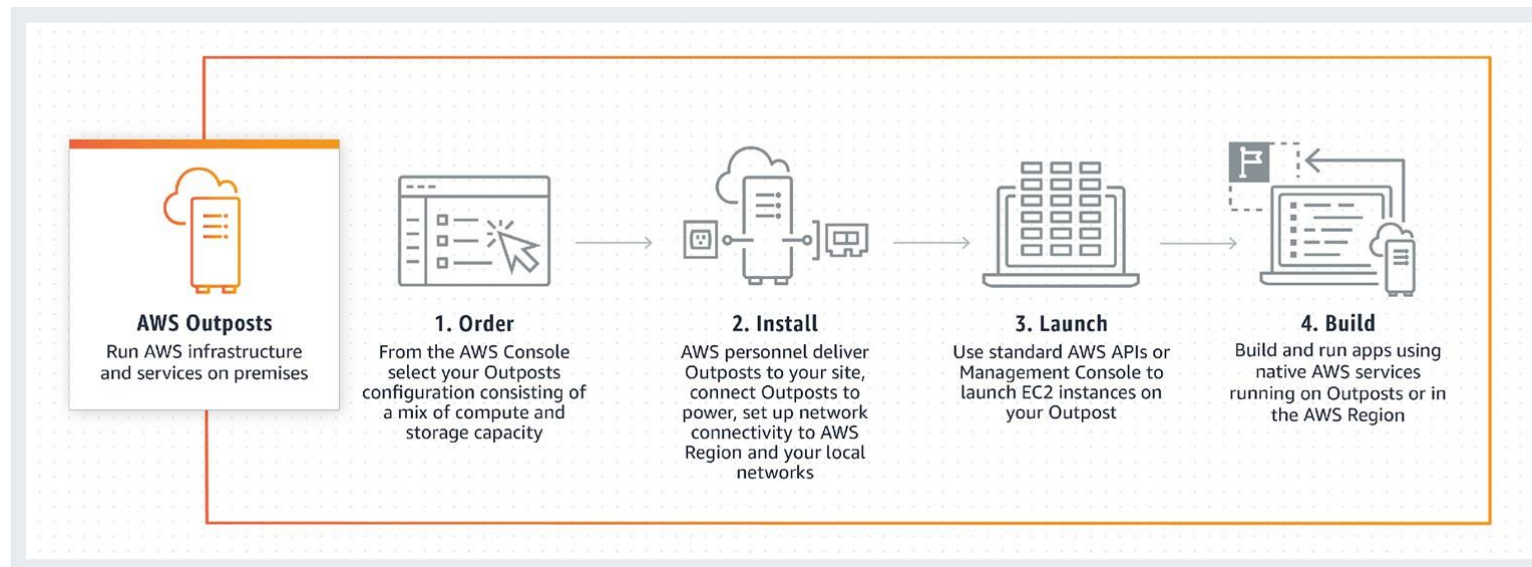


PRIVATE CLOUD

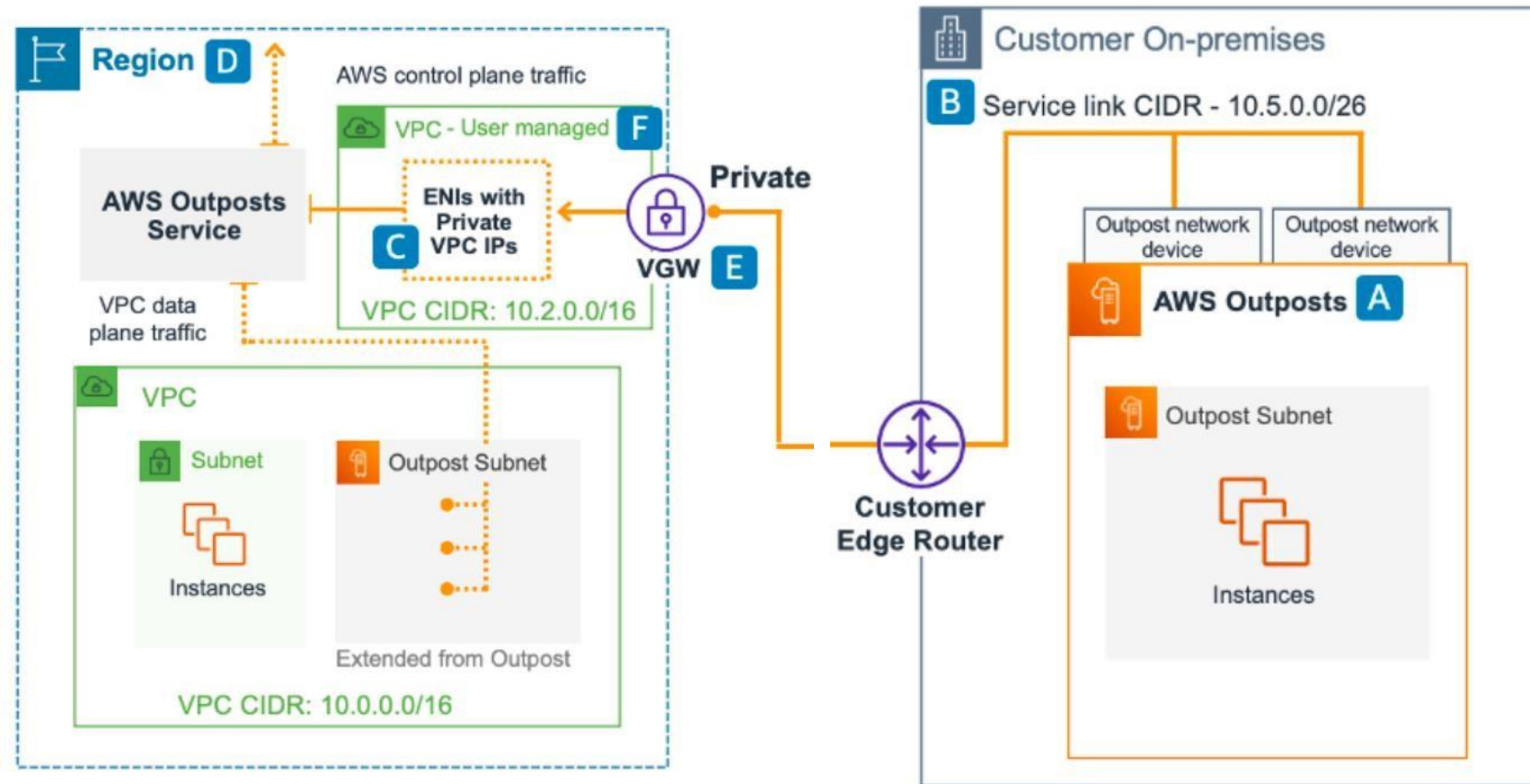
- Offered to select users over the internet or a private internal network
- Provides greater security controls
- Requires traditional data center staffing & maintenance

Private: AWS Outpost

AWS Outposts is a fully managed service that offers the same AWS infrastructure, AWS services, APIs, and tools to virtually any datacenter, co-location space, or on-premises facility for a truly consistent hybrid experience. AWS Outposts is ideal for workloads that require low latency access to on-premises systems, local data processing, data residency, and migration of applications with local system interdependencies.

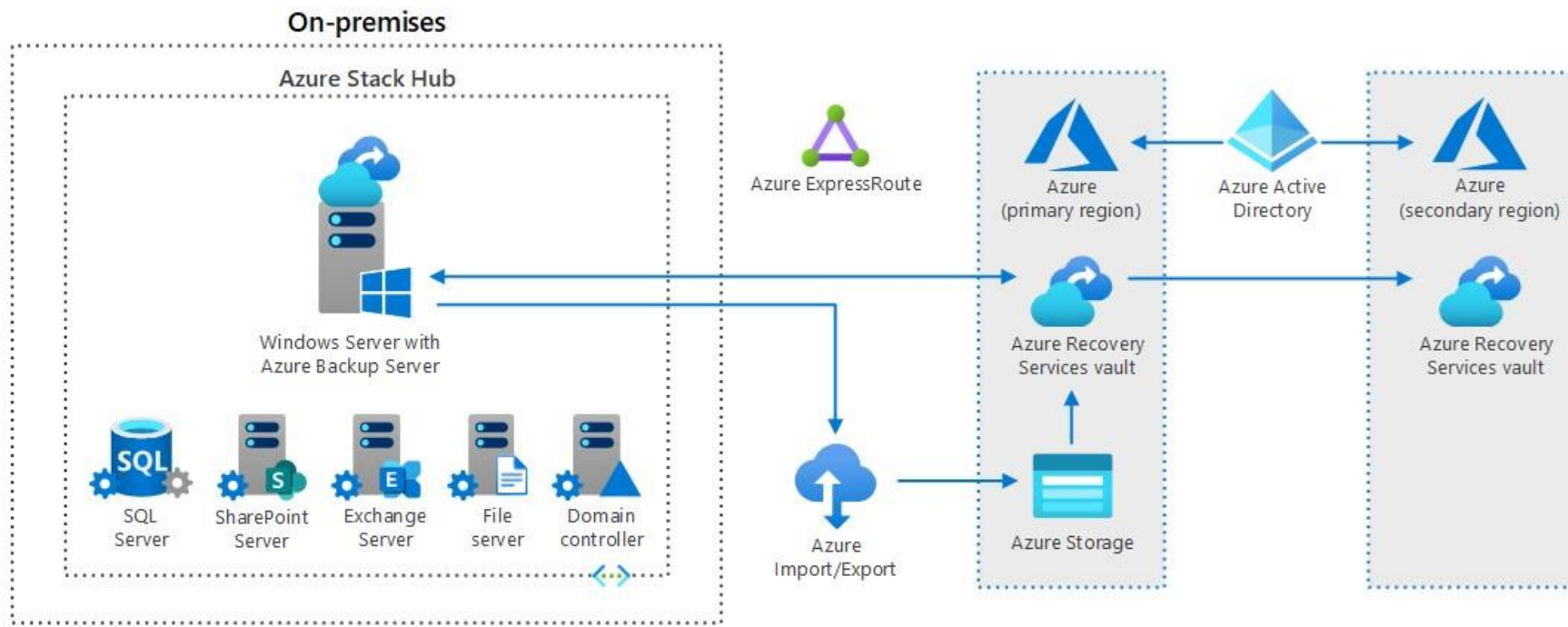


Hybrid Cloud: AWS Outpost



Example: Hybrid Cloud (Azure)

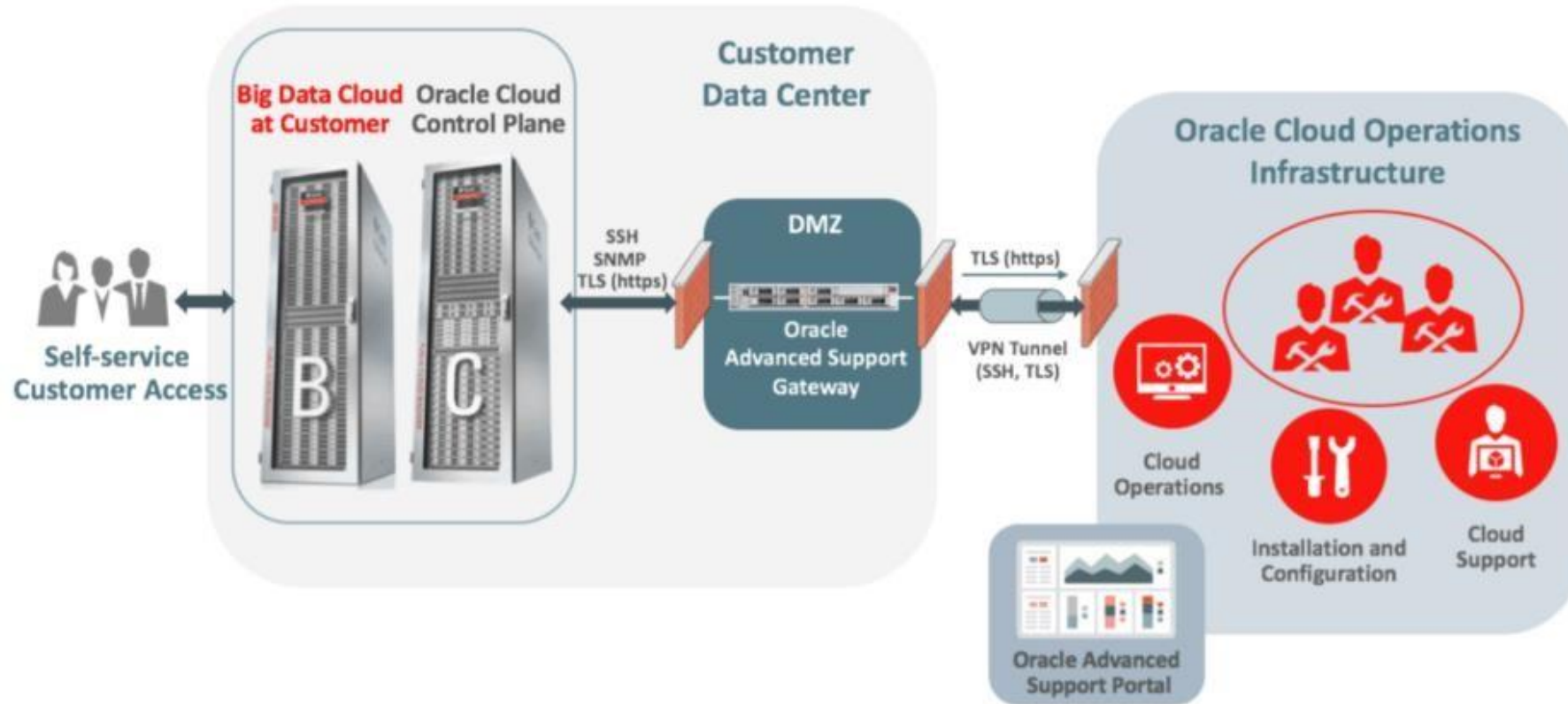
Azure Stack Hub is an extension of Azure that provides a way to run apps in an on-premises environment and deliver Azure services in your datacenter. With a consistent cloud platform, organizations can confidently make technology decisions based on business requirements, rather than business decisions based on technology limitations.



Example: Hybrid Cloud (Google)

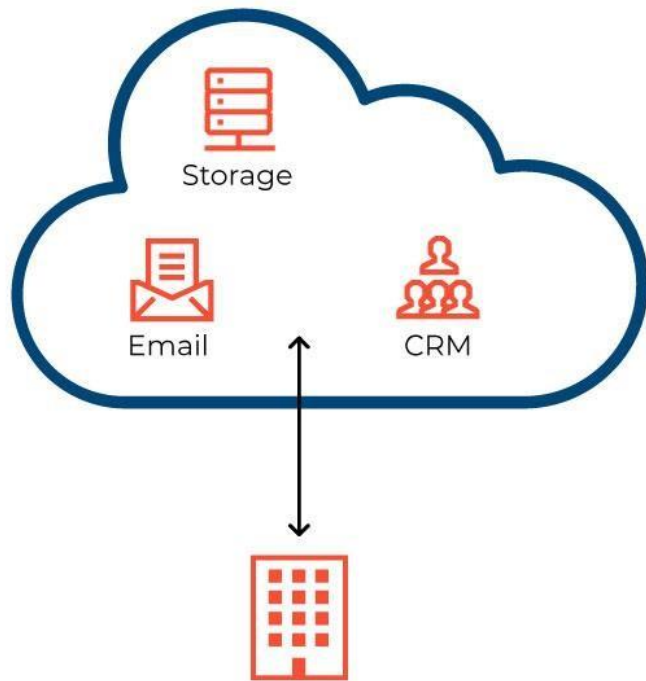


Hybrid Cloud: OCI & C@C



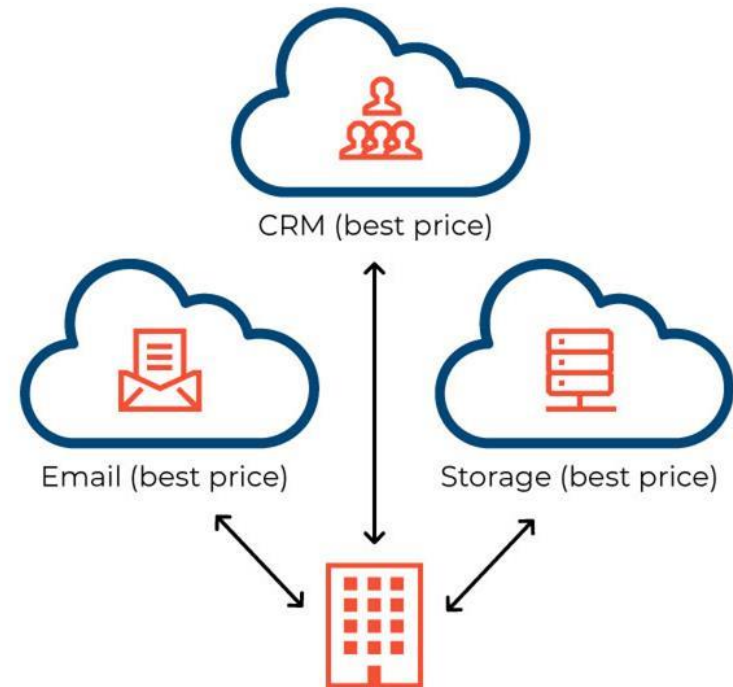
Single vs Multi Cloud

Single cloud



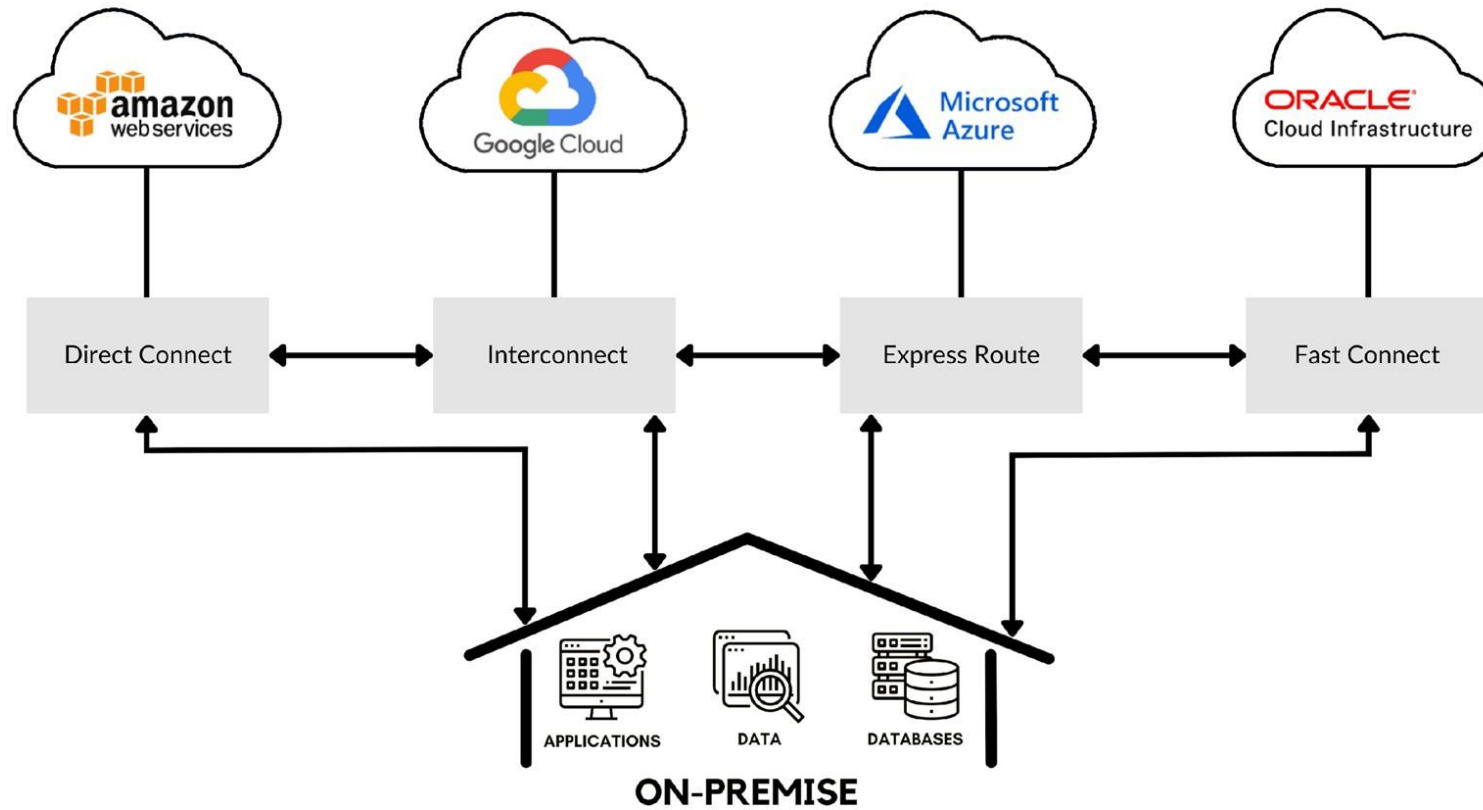
Company is locked in to one cloud vendor

Multi-cloud



Company has more flexibility with multiple cloud vendors

Multi Cloud



Which Cloud?

Cloud Vendors (IaaS & PaaS)

Gartner®

Magic Quadrant for Cloud Infrastructure and Platform Services 2022



Cloud Vendors

Mega-Scale Vendors

ORACLE®

 Google Cloud Platform

 **amazon**
web services

 **IBM**

 Microsoft Azure

 **Alibaba Cloud**
aliyun.com

Which Cloud?

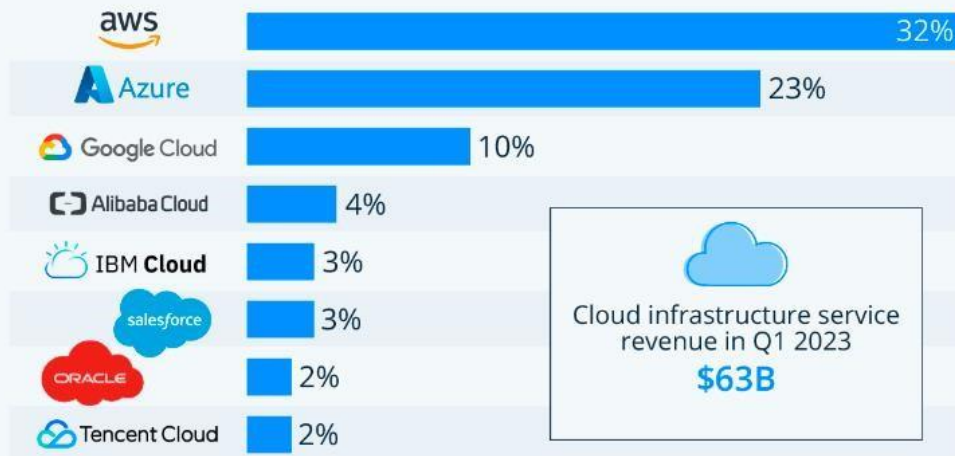


RIGHT



Big Three Dominate the Global Cloud Market

Worldwide market share of leading cloud infrastructure service providers in Q1 2023*



* includes platform as a service (PaaS) and infrastructure as a service (IaaS) as well as hosted private cloud services

Source: Synergy Research Group



statista

AWS Named as a Leader in the 2022 Gartner Cloud Infrastructure & Platform Services (CIPS) Magic Quadrant for the 12th Consecutive Year

Figure 1: Magic Quadrant for Cloud Infrastructure and Platform Services





>90%
of Fortune 500
use Microsoft
Cloud

Amazon Web Services vs Microsoft Azure vs Google Cloud Platform vs Oracle Cloud Infrastructure

Based on verified reviews from real users in the Cloud Infrastructure and Platform Services (Transitioning to Strategic Cloud Platform Services) market. Amazon Web Services has a rating of 4.5 stars with 3539 reviews. Microsoft Azure has a rating of 4.4 stars with 2091 reviews. Google Cloud Platform has a rating of 4.4 stars with 1171 reviews. Oracle Cloud Infrastructure has a rating of 4.3 stars with 341 reviews. See side-by-side comparisons of product capabilities, customer experience, pros and cons, and reviewer demographics to find the best fit for your organization.

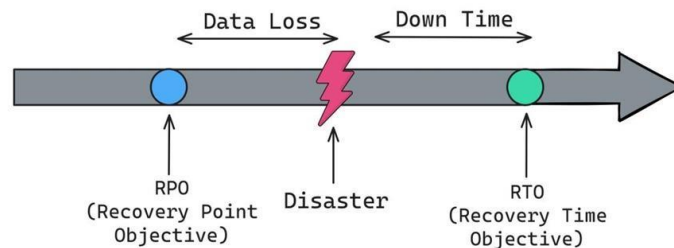


Cloud Comparison

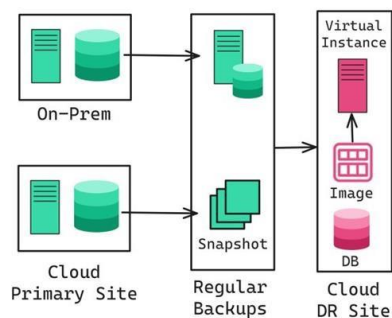
Cloud...

Cloud Disaster Recovery Strategies

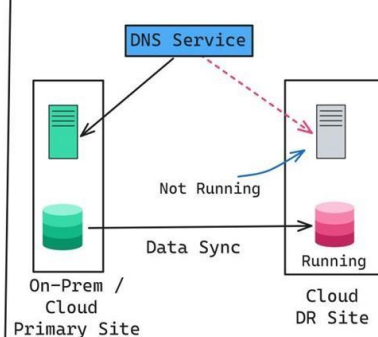
Basics First: What is RTO and RPO ?



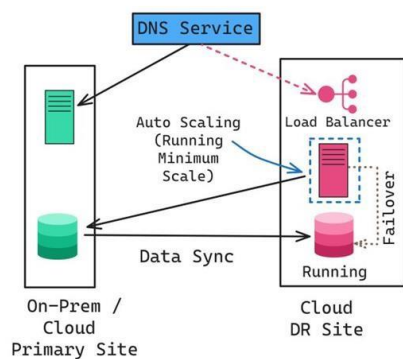
1. Backup and Restore



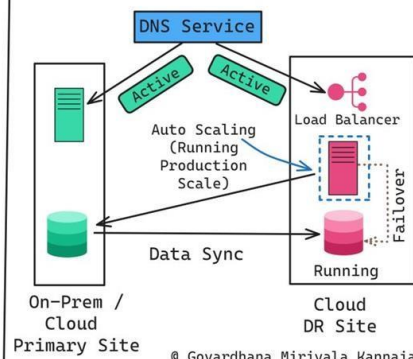
2. Pilot Light



3. Warm Standby



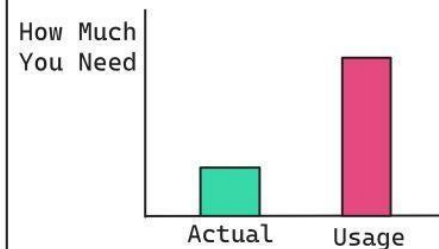
4. Multi Site



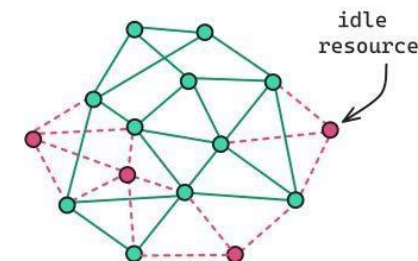
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Cloud Cost Reduction Techniques

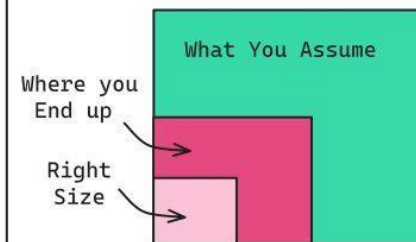
1. Reduce Usage



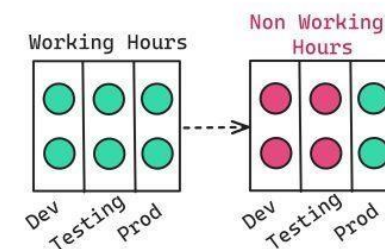
2. Terminate Idle Resources



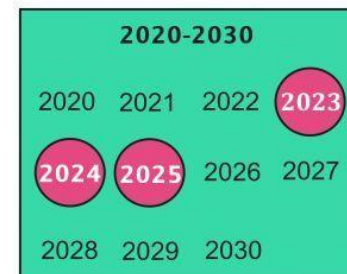
3. Right Sizing



4. Shutdown Resources During Inactive Periods



5. Reserve to Reduce Rate



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6. Optimize Data Transfers

